

Phở

Northern-Southern cross (70/30) · strong, rich, natural beef sweetness · after Leighton Pho, 2024 Science Edition

YIELD	ASSEMBLY	BONES	FINISH
3 L · 6 bowls	30–45 min	1:1 + 30% water	8.7 Brix

BRIX CHECKPOINTS

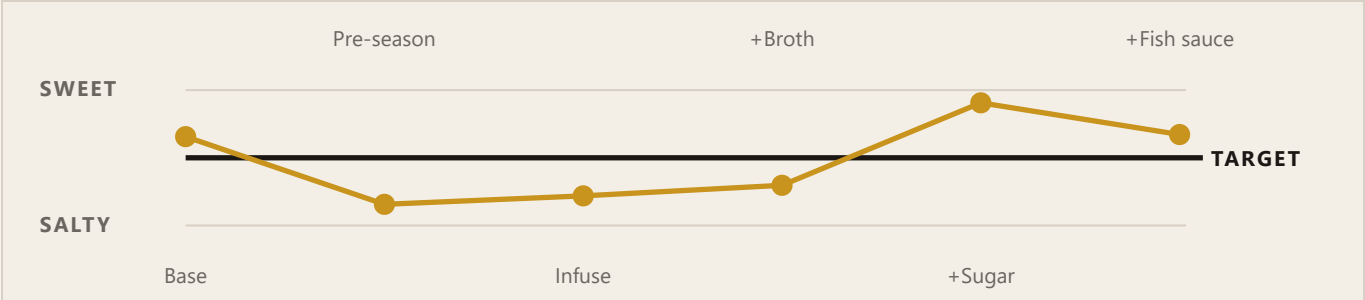
refractometer QC · read a cooled drop, not boiling broth · low → reduce, high → dilute

CHECKPOINT	TARGET	UNDER → REDUCE	OVER → ADD WATER
Base broth — unseasoned, post-strain	7.0%	Simmer down to concentrate up to 7.0 (confirms a full 24-hr extraction)	Top up with water to 3 L @ 7.0
Finished phở — after the fish sauce	8.7%	Reduce the pot a little more, then re-check	Add boiling water to the pot → back to a near-boil → serve all

The blend method: make the bone broth in advance (a weekend project), store it in the fridge or freezer, then transform it into finished phở any weeknight in **30–45 min** — long extraction separated from quick assembly. **Dilution principle:** extract hard, season to ratio, dilute to your 8.7 target. A concentrated base diluted down keeps full flavor and body; an under-extracted broth is missing compounds you *can't* add back.

THE SEASONING CURVE

season in waves — go under, overshoot, then balance to target



Pre-season deliberately salty to find the floor (salt pulls the 24-hr bone flavor forward); **rock sugar** overshoots sweet; **fish sauce** settles it back to the 8.7 target while adding umami.

WITH OR WITHOUT BRISKET

the one branch in the method

	WITH BRISKET	WITHOUT BRISKET
Simmer	~3 hours	30–45 min
Doneness	Probe-tender, not by the clock	Spice / aromatic infusion only
Intensity	Moderate — the meat absorbs broth	More concentrated
At service	Ice-bath → slice thin	Raw steak (tái) in the hot broth

Bone Broth Base

The base is everything. Use **marrow bones** (flavor) + **knuckle bones** (gelatin). The ratio is **1:1 bones to final volume** — add **30% extra water** to cover evaporation over the long simmer. This is the weekend project; the weeknight assembly (page 3) is fast because this is done.

BONE ROASTING

reserve the rendered fat — it's your tallow for the spice bloom

- 1 **Straight from frozen:** bones go from freezer to oven — **no thaw, no parboil.**
- 2 **Roast hot: 425–450°F for ~1 hour.**
- 3 **Color target: golden-mahogany with dark spots** — dark spots are fine, black is not.
- 4 **Save the fat:** reserve the rendered tallow for blooming the spices at assembly.

COLOR & CHAR CUES

judged by eye — over-charring turns it bitter

Bone color	Golden-mahogany. Dark spots OK, not black.
Aromatics char	30–40% black spots. Not full char — full char causes bitterness.

24-HOUR SIMMER

gentle bubbles, never a rolling boil — that's what keeps it clear

- 1 **Set it:** roasted bones + water (the 30% extra for evaporation).
- 2 **Hold the heat:** simmer at **~94°C / 200°F** — gentle bubbles, **no rolling boil** (a hard boil emulsifies fat and turns it murky).
- 3 **Finish to number:** top up to the target volume at the end, strain. **Target: 7.0 Brix.**
- 4 **Store:** fridge up to **1 week**, or freezer for **months.**

Body comes from gelatin, not luck. If the chilled broth doesn't set to a wobble, you're short on knuckle bones or the simmer was cut early — the full 24 hr at a gentle 200°F is what converts collagen to the gelatin that gives phở its weight on the tongue.

Weeknight Assembly

INGREDIENTS

Base: 3 L bone broth @ 7.0 Brix (from page 2)

AROMATICS

Brown onion	½
Shallots halved	2–3
Ginger skin on	1 knob
Garlic cloves	4–5

MEAT

Brisket optional	500 g
— or raw steak for tái	—

SPICES

Cinnamon cassia	12 g
Black cardamom	14 g
Star anise	20 g
Clove	2 g

X-FACTOR

Oxtail concentrate optional, adds depth	250 ml
---	--------

PRE-SEASONING

Salt 1.17%	35 g
Rock sugar 1.83%	55 g
Hạt nêm / LSP 0.67%	20 g

FINAL ADJUSTMENT

Fish sauce to 8.7 Brix	8–15 ml
------------------------	---------

METHOD

bloom in the reserved tallow off-heat — the broth quenches it

- Char aromatics:** high broil — onion halves, shallots, ginger (skin on), garlic. Flip for even char. Target **30–40% black spots**, not full char (bitter).
- Bloom spices:** beef tallow in the pot, get it hot, **kill the heat**. Add the spices ~30 sec until fragrant, then the charred aromatics — they sizzle in the residual heat.
- Add broth + pre-season:** pour in the **3 L bone broth** (it quenches everything), bring to a gentle simmer. Add salt (35 g), rock sugar (55 g), hạt nêm (20 g); stir to dissolve. Taste — 95% there, slightly salty.
- Infuse:** without brisket **30–45 min**; with brisket **~3 hr** to probe-tender. Maintain the volume — big loss means the heat's too high.
- Final adjustment:** strain out the aromatics + spices. Add **fish sauce (8–15 ml)** until you catch a slight fishiness; optional 250 ml oxtail concentrate. **Target 8.7 Brix** — if over, add boiling water to the pot and bring back to a near-boil before serving.
- Serve:** noodles in the bowl, ladle hot broth, add sliced brisket or raw steak (it cooks in the broth). Sides: bean sprouts, Thai basil, lime, hoisin, sriracha.

TROUBLESHOOTING

Northern style: ginger + onion + 1 star anise, sugar cut 50%+, fish sauce as finishing salt, dilute harder

PROBLEM	CAUSE	FIX
Too intense	Over-extraction / no brisket	Dilute to 8.7 Brix at service
Too weak	Under-extraction (under 7 Brix base)	Simmer longer or reduce
Bitter	Burnt cloves / over-charred aromatics	Lower heat; 30–40% char max
Murky	Boiled too hard	Gentle simmer; skim early
No body	Low gelatin extraction	Use knuckle bones; full 24 hr